Demo: Loading Macros via include shortcode

This page demonstrates the `include` shortcode strategy for loading macros into a Quarto document.

The `{{< include macros.qmd >}}` shortcode at the top of this file embeds the entire contents of `macros.qmd` inline at that point. Since `macros.qmd` contains only raw LaTeX macro definitions (`\def`, `\providecommand`, etc.), Quarto passes them through as raw LaTeX for PDF output (where they are processed inline in the document body as LaTeX commands) and as math for HTML/RevealJS output (where they are processed by MathJax).

> **Note:** This demo uses `macros.qmd` directly (same directory). When using this repository as a submodule, use `macros/macros.qmd` instead.

> **Note (MathJax limitations):** Quarto's MathJax integration supports `\def`, `\newcommand`, `\renewcommand`, `\newenvironment`, `\renewenvironment`, and `\let` — but **not** `\providecommand`. Since `macros.qmd` uses `\providecommand` for many definitions, those macros are silently ignored for HTML/RevealJS output. The example table below uses only `\def`-based macros, which render correctly in all formats.

Proof that `\providecommand` is ignored in HTML/RevealJS

The macro `\floor` in `macros.qmd` is defined with `\providecommand`. Compare this with a `\def` macro:

Macro type	Code	Output
`\providecommand` macro (`\floor`)	`$\floor{1.2}$`	$\lfloor 1.2 \rfloor$
`\def` macro (`\a`)	`$\a$`	α

In HTML/RevealJS, `\floor` remains unresolved while `\a` renders correctly, showing that `\providecommand` definitions are ignored by MathJax.

Note (RevealJS blank slide): Content placed before the first `##` heading in RevealJS output becomes a blank title slide. Since the include shortcode is at the top of this file, RevealJS may show a blank slide before the first section. This is a known limitation of the shortcode approach for RevealJS. For RevealJS without a blank slide, use the `include-in-header` strategy instead.

Example math

The table below uses macros defined with `\def` in `macros.qmd`. These `\def`-based macros work across all output formats — note that `\providecommand`-based macros from `macros.qmd` are silently ignored for HTML/RevealJS as noted above:

Description	Code	Output
Greek: alpha	`$\a$`	α
Greek: beta	`$\b$`	β
Greek: gamma	`$\g$`	γ
Greek: lambda	`$\l$`	λ
Greek: sigma	`$\s$`	σ
Log-likelihood	`$\llik$`	ℓ
Independent	`$X \ind Y$`	$X \perp\!\!\!\perp Y$
IID distributed	`$X_i \siid \Normal(\m, \ss)$`	$X_i \ \sim_{\text{iid}}\ \mathrm{N}(\mu, \sigma^2)$
Vector: mu	`$\vm$`	$\tilde{\mu}$
Hat beta	`$\hb$`	$\hat{\beta}$